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# React useState Hook

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The React `useState` Hook allows us to track state in a function component.

State generally refers to data or properties that need to be tracking in an application.

## Import `useState`

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To use the `useState` Hook, we first need to `import` it into our component.

### Example:

Get your own React.js Server

At the top of your component, `import` the `useState` Hook.

```
import { useState } from "react";
```

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To learn more about destructuring, check out the ES6 Destructuring section.

# Initialize `useState`

We initialize our state by calling `useState` in our function component.

## Example:

Initialize state at the top of the function component.

```
import { useState } from "react";

function FavoriteColor() {
  const [color, setColor] = useState("red");
}
```

Notice that again, we are destructuring the returned values from `useState`.

The first value, `color`, is our current state.

The second value, `setColor`, is the function that is used to update our state.

These names are variables that can be named anything you would like.

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# Read State

We can now include our state anywhere in our component.

## Example:

Use the state variable in the rendered component.

```
import { useState } from 'react';
import { createRoot } from 'react-dom/client';

function FavoriteColor() {
  const [color, setColor] = useState("red");

  return <h1>My favorite color is {color}!</h1>
}

createRoot(document.getElementById('root')).render(
```

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# Update State

To update our state, we use our state updater function.

## Example:

Use the state updater function to update the state:

```
<button
  type="button"
  onClick={() => setColor("blue")}
>Blue</button>
```

We should never directly update state. Ex: `color = "blue"` is not allowed.

## Example:

Use a button to update the state:

```
import { useState } from 'react';
import { createRoot } from 'react-dom/client';

function FavoriteColor() {
  const [color, setColor] = useState("red");
```

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```
        type="button"
        onClick={() => setColor("blue")}
      >Blue</button>
    </>
  )
}

createRoot(document.getElementById('root')).render(
  <FavoriteColor />
);
```

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# What Can State Hold?

The `useState` Hook can be used to keep track of strings, numbers, booleans, arrays, objects, and any combination of these!

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Example:

Create multiple state Hooks:

```
import { useState } from 'react';
import { createRoot } from 'react-dom/client';

function MyCar() {
  const [brand, setBrand] = useState("Ford");
  const [model, setModel] = useState("Mustang");
  const [year, setYear] = useState("1964");
  const [color, setColor] = useState("red");

  return (
    <>
      <h1>My {brand}</h1>
      <p>
        It is a {color} {model} from {year}.
      </p>
    </>
  )
}

createRoot(document.getElementById('root')).render(
  <MyCar />
);
```

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Or, we can just use one state and include an object instead!

## Example:

Create a single Hook that holds an object:

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```
function MyCar() {
  const [car, setCar] = useState({
    brand: "Ford",
    model: "Mustang",
    year: "1964",
    color: "red"
  });

  return (
    <>
      <h1>My {car.brand}</h1>
      <p>
        It is a {car.color} {car.model} from {car.year}.
      </p>
    </>
  )
}

createRoot(document.getElementById('root')).render(
  <MyCar />
);
```

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Since we are now tracking a single object: `car`, we need to reference that object when rendering the component. (Ex: `car.brand`)

# Updating Objects and Arrays in State

When state is updated, the entire state gets overwritten.

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We can use the JavaScript spread operator to help us.

## Example:

Use the JavaScript spread operator to update only the color of the car:

```
const updateColor = () => {
  setCar(previousState => {
    return { ...previousState, color: "blue" }
  });
}
```

Run Example »

Because we need the current value of state, we pass a function into our `setCar` function. This function receives the previous value.

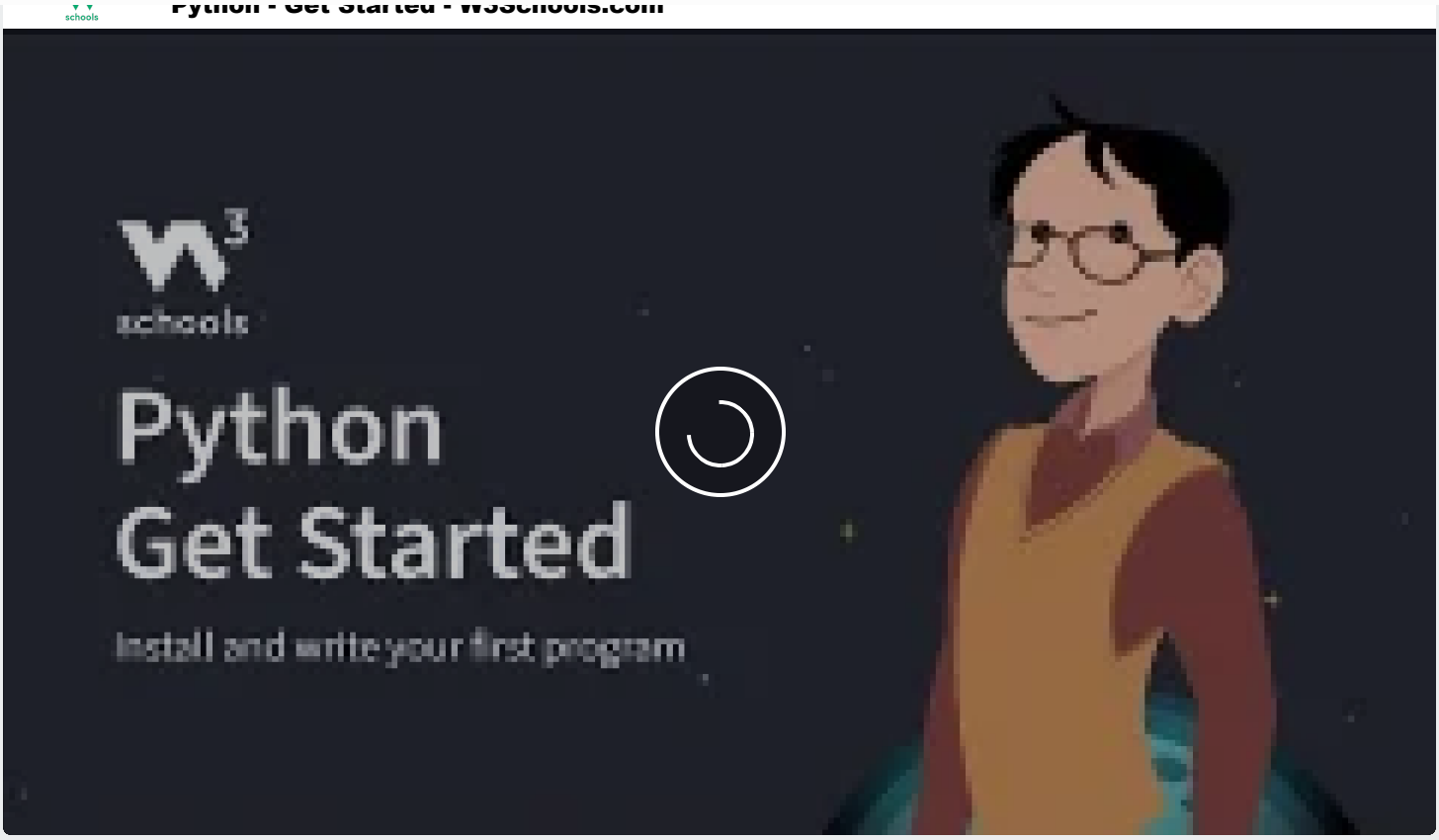
We then return an object, spreading the `previousState` and overwriting only the color.

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